

Cost-Aware Evaluation of Time-Series Foundation Models for Urban Air-Quality and Temperature Forecasting

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ABSTRACT

Cities with the greatest need for hyper-local air-quality and weather forecasts often have the shortest data records and the tightest compute budgets. The usual responses, searching for a tiny architecture or transferring from data-rich cities, assume foundation models are too heavy for such deployments, an assumption more often adopted than tested. Across 29 cities and two domains (hourly PM2.5 and 2 m temperature), under pre-specified protocols with measured energy and a cost-adjusted decision rule, we compare all three. On PM2.5 three zero-shot foundation models spanning two families and a 4.3-fold size range are statistically indistinguishable from the tuned specialist (MASE 0.659-0.662 versus 0.692 under deployable causal covariates) and formally equivalent to one another. The specialist's apparent temperature edge (0.533 versus 0.792) depends heavily on covariate timing: 0.638 with covariates carrying measured 24-hour forecast error, 0.745 with only values knowable at the forecast origin. Perfect foresight therefore distorts the cross-domain comparison. Any benchmark using future covariates carries the same risk. Transfer learning never significantly beats zero-shot at any budget. Under frequent retraining the untrained model uses roughly ten times less measured energy, though a once-trained specialist eventually amortizes. Where models are refreshed often, downloading one now matches training one.

Keywords: time-series foundation models, air-quality forecasting, temperature forecasting, neural architecture search, transfer learning, edge computing, energy-aware machine learning, deployment cost

1 Introduction

Ambient fine particulate matter is among the largest environmental risk factors for human health, responsible for millions of premature deaths annually and concentrated disproportionately in cities of low- and middle-income countries^{1,2}. Managing that burden (issuing health advisories, timing traffic or industrial interventions, planning sensor maintenance) requires hyper-local, short-horizon forecasts at the level of an individual monitor. The cities that need such forecasts most are also those where monitoring is newest and sparsest. Low-cost sensor networks and open platforms such as OpenAQ have widened coverage^{3,4}, but the records they produce are short, gappy, and heterogeneous, and the institutions operating them rarely command cloud budgets or ML engineering teams⁵. The same constraint binds for local weather variables such as near-surface temperature, whose station-level forecasts feed heat-warning and energy systems⁶: global deep-learning weather models now rival numerical weather prediction (NWP)^{7,8}, but their outputs and compute remain out of reach of a single-station operator on an edge budget.

A resource-constrained operator today faces three deployment strategies. The first is to train an efficient specialist locally. Gradient-boosted trees remain a strong widely deployable baseline for tabularized forecasting⁹⁻¹¹, deep specialists from recurrent probabilistic models to graph and transformer architectures are widely represented in the air-quality and general forecasting literatures¹²⁻¹⁹, and neural architecture search (NAS) refines the recipe by looking for an architecture that meets an explicit efficiency budget²⁰⁻²². The second is transfer learning: pretrain where data are plentiful, then fine-tune on the short local record²³. In our prior conference study we combined the two routes: we discovered a 153k-parameter recurrent forecaster by multi-objective search and showed that transfer from data-rich cities recovers most of the accuracy lost to short local histories²⁴. The present work inherits that architecture and its edge/energy framing, and adds the foundation-model tier, the cost-adjusted decision rule, the air-quality domain, and the transfer-versus-zero-shot crux experiment. The third strategy, available only since 2023, is to train nothing at all: pretrained time-series foundation models (TSFMs) such as Chronos, TimesFM, Moirai, Lag-Llama and TimeGPT forecast zero-shot from the local context window alone²⁵⁻³¹.

Which strategy should such an operator choose? The benchmark literature does not settle the question. Foundation-model evaluations report aggregate leaderboards across heterogeneous datasets³², and recent critiques question how robust the

24 zero-shot advantage is³³; the doubt echoes much older evidence that complex methods do not automatically beat simple ones³⁴.
25 Deployment cost is largely absent from these comparisons, although the green-AI literature has long argued that energy belongs
26 in the objective^{35–39} and the edge-ML community treats the joule budget as the binding constraint^{40,41}. Underlying much of
27 that work, including our conference study above²⁴, is the assumption that foundation models are too heavy for the edge and that
28 efficiency must therefore come from search and transfer. The assumption is reasonable, and it has been adopted far more often
29 than it has been tested. Compact foundation models make the test feasible: Chronos-Bolt-small runs on a consumer GPU⁴², so
30 what was a design premise becomes a question the data can answer.

31 Three gaps follow from that literature, and this study addresses each. First, foundation models are assessed as aggregate
32 leaderboards over heterogeneous collections^{32,33}, which establishes whether a model is good on average but not which of
33 the three strategies a particular operator should deploy under a particular budget. Second, where deployment cost appears at
34 all it is a parameter count or an inference latency rather than measured energy, so the green-AI position that energy belongs
35 in the objective^{35–37} has not been carried through into a model-selection rule an operator could apply. Third, the air-quality
36 forecasting literature and the foundation-model literature have developed largely apart: the former tunes specialists on one
37 city or one country^{12,14,15}, the latter evaluates general-purpose models on curated archives, and neither speaks to the short,
38 gappy record of the data-scarce city that motivates this work. Abbreviations and symbols used throughout are collected in
39 Supplementary Table S1.

40 A methodological problem compounds the strategy question. Forecast evaluation has well-documented pitfalls^{43,44}, and we
41 examine one that matters specifically for comparisons across domains. Standard evaluation harnesses hand covariate-using
42 models their exogenous inputs at the forecast target time (the `future_covariates` convention), which quietly assumes
43 a perfect weather forecast over the horizon. The assumption is harmless where covariates carry little information about the
44 target and decisive where they are near-deterministic drivers of it, and we show below that it can, by itself, produce a spurious
45 domain-level conclusion about which strategy wins.

46 Here we test the three strategies directly. We compare five tiers on a 29-city panel (14 data-rich, 15 data-scarce) in two
47 domains, urban PM2.5 and location-level temperature: a seasonal-naïve floor, a LightGBM specialist, the NAS-discovered
48 Green-NAS-A model, zero-shot Chronos-Bolt, and a covariate-residual Chronos variant. Evaluation uses pre-specified quality
49 gates, six-fold rolling-origin backtesting, and Diebold–Mariano, sign, and Wilcoxon tests under false-discovery-rate (FDR)
50 control, with paired equivalence tests wherever we claim a tie; uncertainty is quantified with split-conformal intervals, and
51 energy is measured directly on edge/CPU-class hardware (a consumer GPU and a CPU-only configuration). A pre-specified
52 crux experiment (E4) pits the transfer-learning recipe against the zero-shot model across fine-tune budgets on the scarce cities.
53 A cost-adjusted decision rule, minimizing the mean absolute scaled error (MASE) + $\lambda \cdot \text{USD}$ per 1,000 forecasts with λ a
54 cost-penalty coefficient, then converts the results into deployment winner maps.

55 The paper makes three contributions.

56 The first is the test that the premise had not received. To our knowledge this is the first head-to-head comparison of search,
57 transfer, and zero-shot deployment for hyper-local urban forecasting in which accuracy, measured energy, and cost enter a
58 single decision rule, on a panel large enough to separate the data-rich case from the data-scarce one. The premise does not
59 survive it. On PM2.5, zero-shot Chronos-Bolt and the tuned specialist are statistically indistinguishable across the panel: neither
60 records an FDR-significant win over the other, a paired two-one-sided equivalence test (TOST) declares the two equivalent at
61 a 0.05-MASE margin when the specialist is granted perfect-foresight covariates, and under deployable causal covariates the
62 foundation model is, if anything, ahead. This is not a property of one checkpoint: two further zero-shot models reproduce it,
63 one changing checkpoint scale 4.3-fold within the same family and one changing family at matched scale, and all three are
64 certified equivalent to one another at the same margin. Its advantage shows no detectable association with local data volume. In
65 the data-scarce regime, transfer learning never significantly beats the zero-shot model at any fine-tune budget, and the zero-shot
66 model leads at every one.

67 The second is methodological and outlives this dataset: covariate timing alone can distort a domain-level conclusion. On
68 temperature, most of the specialist’s large apparent superiority is an artifact of the standard future-covariates convention. We
69 quantify how much by calibrating an intermediate regime against measured 24-hour weather forecast error: supplied with
70 covariates of that realistic quality the specialist keeps roughly three-fifths of its perfect-foresight advantage, and restricted to
71 values knowable at the forecast origin it keeps under a fifth. The convention therefore inflates a cross-domain contrast that a
72 realistically equipped specialist only partly supports, and the corrected picture is graded rather than domain-flipped. Because
73 the distortion scales with how informative the covariates are about the target, it is nearly invisible within one domain and
74 surfaces only in comparisons across domains, which is exactly where it does the most damage. Any benchmark that consumes
75 future covariates and compares across domains needs a causal-covariate ablation to rule it out.

76 The third is the deployment consequence, which is what an operator can act on. Under a frequent-retraining protocol
77 the zero-shot model uses roughly an order of magnitude less measured energy than the tiers that train locally. Separating
78 training from inference shows that gap is training-driven rather than intrinsic: a once-trained specialist eventually amortizes and

becomes the more energy-efficient option, so the defensible recommendation is conditional on refresh cadence rather than universal. For a city that retrains often, lacks a trustworthy forecast of its own covariates, or has no ML engineering capacity to spare, the practical default has moved from training a small model to downloading one. We release the decision rule so that an operator can re-derive that answer at their own electricity price, hardware, and retraining frequency.

Results

Study design

Five model tiers in two domains, urban PM2.5 (primary) and 2 m air temperature (secondary), on one 29-city panel under one pre-specified harness (Fig. 1; Methods). The tiers span the three options open to a resource-constrained operator: *search* (NAS-GRU, the architecture carried over from our earlier conference study²⁴), *specialize* (LightGBM with weather and calendar covariates⁹), and *train nothing* (Chronos-Bolt-small run zero-shot, with and without a covariate-residual variant^{25,42}), plus a seasonal-naïve floor. Accuracy is MASE on a fixed per-series scale; energy is measured end to end⁴⁵. Results below are at the 24-hour horizon; the 48-hour horizon behaves the same way (Supplementary Table S9). Abbreviations used throughout are collected in Supplementary Table S1.

A zero-shot foundation model matches the tuned specialist on PM2.5

On the 29-city PM2.5 panel, zero-shot Chronos-Bolt and the tuned LightGBM specialist were statistically indistinguishable. In the deployable causal-covariate configuration reported in Table 1, panel-mean MASE is 0.662 ± 0.368 (Chronos-Bolt) versus 0.692 ± 0.374 (specialist) (Fig. 2a); with perfect-foresight covariates the specialist matches Chronos-Bolt exactly (0.662, Supplementary Table S13). Neither model recorded a single FDR-corrected DM-significant win over the other across all 29 cities, in either the causal panel (0/0) or the perfect-foresight panel (0/0) (full perfect-foresight pairwise matrices in Supplementary Table S5). In the perfect-foresight configuration the panel-level tests were non-significant (sign test $P = 0.136$, Wilcoxon signed-rank $P = 0.325$). Under causal covariates the per-city balance in fact tilts toward the foundation model, which is better in 21 of 29 cities (sign test $P = 0.024$), but the magnitude-weighted test does not reach significance (Wilcoxon $P = 0.084$). We therefore describe the deployable comparison conservatively, as no significant difference rather than a foundation-model win.

A non-significant test is not evidence of equivalence, so we tested equivalence directly. Under perfect foresight the agreement clears that stricter bar: a paired two-one-sided test against a margin of $\delta = 0.05$ MASE declares the perfect-foresight specialist *equivalent* to Chronos-Bolt (TOST $P = 0.026$; the 90% interval that TOST requires at $\alpha = 0.05$, $[-0.042, 0.042]$, lies inside the margin). Under causal covariates the point difference is $+0.030$ MASE and non-significant, but the interval is a little too wide to certify equivalence at the same margin (upper bound 0.071), so we report that comparison as “no significant difference” only. All seven equivalence tests appear in Supplementary Fig. S2 and their sensitivity to δ in Supplementary Table S15.

This behaviour is not specific to one architecture or checkpoint. The panel was repeated with two further zero-shot models on identical folds (Methods): Chronos-Bolt-base (205 M parameters), which changes checkpoint scale by 4.3-fold while holding the family fixed, and TimesFM 2.5 (231 M parameters, decoder-only, a different developer and pretraining corpus), which changes family at a scale matched to Chronos-Bolt-base.

Neither axis moves the result. All three models fall within 0.003 MASE of one another (0.659 TimesFM, 0.660 Chronos-Bolt-base, 0.662 Chronos-Bolt-small, against 0.692 for the specialist), and no pair of them records a single FDR-significant DM win in any of the 29 cities; the two 200 M-class models from different developers do not differ significantly in even one city. Paired TOST certifies *equivalence* at $\delta = 0.05$ along both axes separately: scale within a family ($P = 1.0 \times 10^{-5}$) and family at matched scale ($P = 2.5 \times 10^{-5}$), as well as between the original Chronos-Bolt-small and TimesFM ($P = 3.6 \times 10^{-5}$). Per-city MASE correlates at $r = 0.99$. Each model stands in the same relation to the specialist (mean differences $+0.030$ to $+0.033$ MASE, no FDR-significant win in either direction), each clears the seasonal-naïve floor decisively (12-15 FDR-significant wins), and the covariate-residual variant degrades for all three alike (0.780-0.797).

We therefore report the PM2.5 conclusion for compact zero-shot foundation models rather than for Chronos-Bolt alone. The claim rests on three checkpoints spanning two families and a 4.3-fold range of model size, which is not the whole model class: every model evaluated here is compact by foundation-model standards, and none exceeds 231 M parameters.

On temperature the three models do not collapse onto one another as they do on PM2.5, and the same two-axis design localises why. Panel-mean MASE rises from 0.793 (Chronos-Bolt-small) to 0.806 (Chronos-Bolt-base) to 0.829 (TimesFM). Changing checkpoint scale 4.3-fold within the Chronos family is certified *equivalent* (paired difference -0.013 , TOST $P = 7.6 \times 10^{-4}$), whereas changing family at matched scale is not (-0.024 , $P = 0.105$). The divergence in this domain is therefore a property of the architecture family rather than of model size.

That distinction carries into the headline comparison. Against the tuned specialist, the paired panel difference is -0.048 for Chronos-Bolt-small and -0.061 for Chronos-Bolt-base, both with 95% intervals containing zero, but -0.085 for TimesFM,

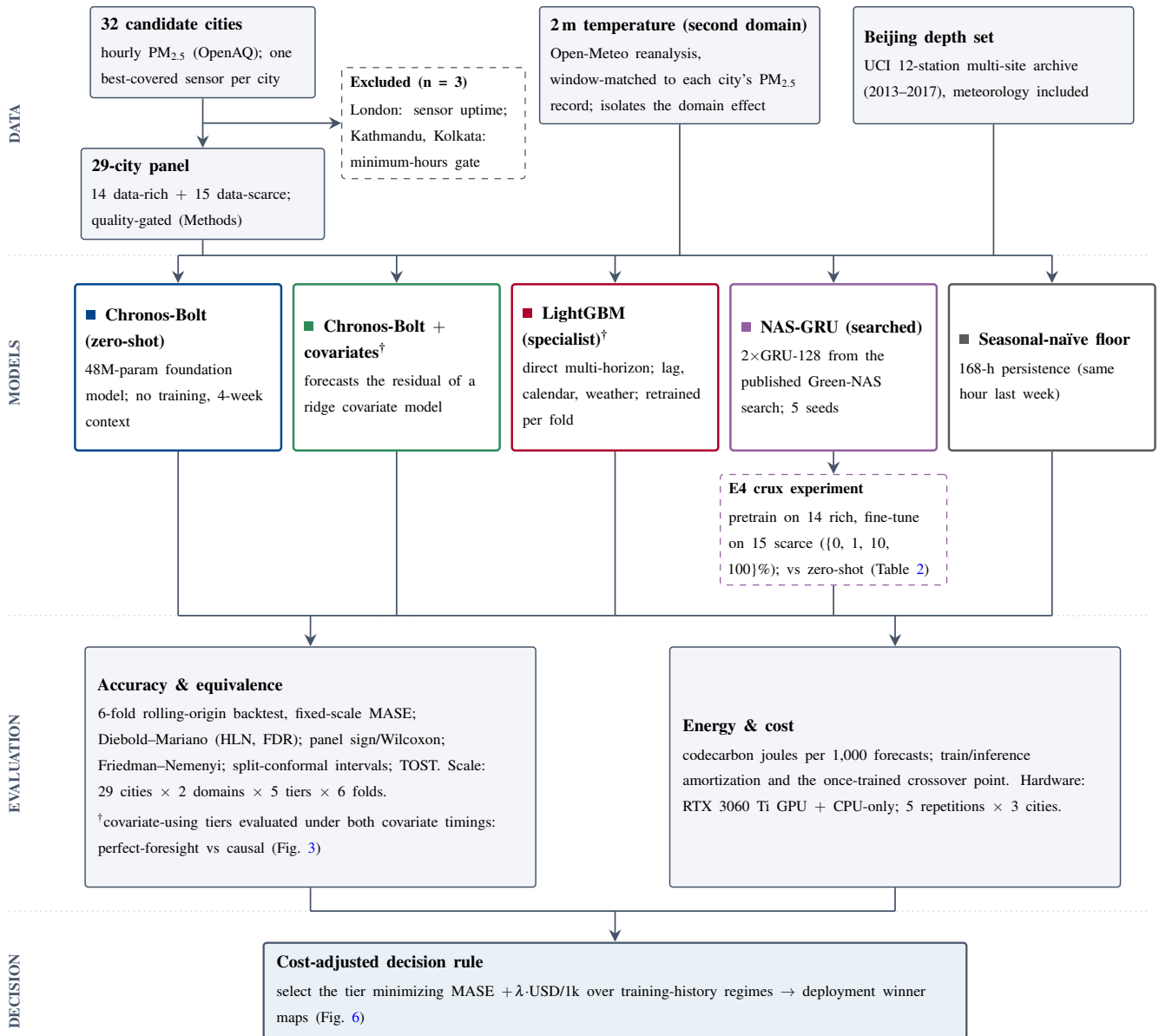


Figure 1. Study design: five forecasting strategies, two domains, 29 cities. Of 32 candidate cities, 3 were excluded under pre-specified quality criteria, leaving a 29-city panel (14 data-rich, 15 data-scarce). Hourly PM_{2.5} (OpenAQ, one reference sensor per city) and 2 m temperature (Open-Meteo, window-matched to each city's PM_{2.5} record) are forecast 24 h ahead by five tiers spanning search (NAS-GRU, with the E4 transfer experiment), specialization (LightGBM direct multi-horizon), zero-shot foundation modelling (Chronos-Bolt-small, with and without a covariate-residual variant), and a seasonal-naïve floor; daggers mark the covariate-consuming tiers evaluated under both covariate timings. Every tier is scored on identical six-fold rolling-origin backtests with fixed-scale MASE, Diebold–Mariano/sign/Wilcoxon inference under FDR control, split-conformal intervals, and TOST equivalence tests, and separately on codecarbon-measured energy; the cost-adjusted decision rule then combines accuracy and cost into deployment winner maps.

whose interval excludes it ($[-0.148, -0.022]$). The per-city Diebold–Mariano tests detect none of this: no FM pair records more than one FDR-significant city, and the specialist records no FDR-significant win over TimesFM in any city. Since the per-city test is the less powerful of the two, we read the panel-level result as the more informative and report both. Our temperature parity claim therefore holds for the Chronos family across a 4.3-fold size range but not for TimesFM. All three clear the seasonal-naïve floor, and the covariate-residual variants degrade alike (2.61 ± 0.01 across the three), so that collapse is a property of the residual scheme under causal covariates rather than of any one model.

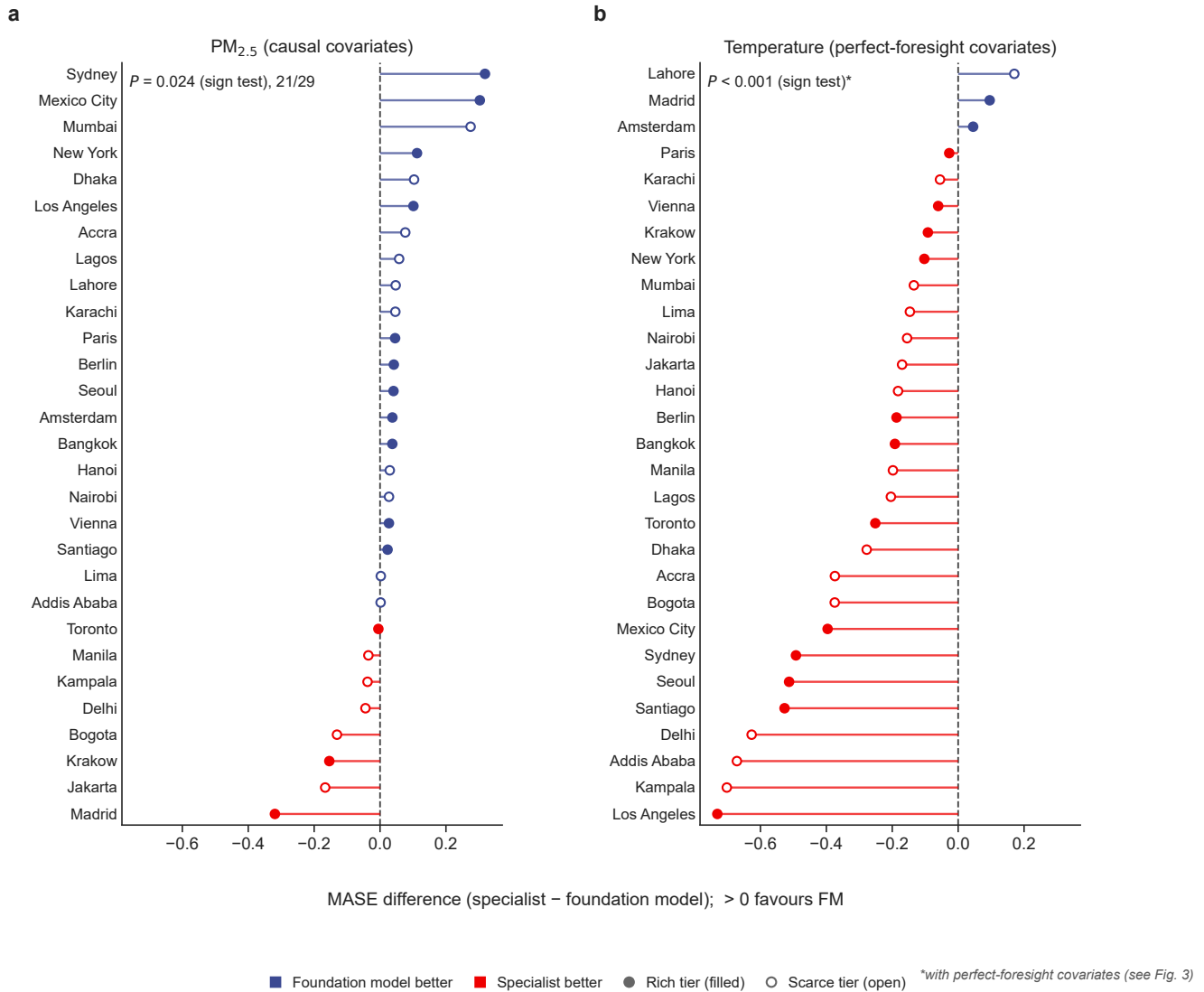


Figure 2. Zero-shot foundation model versus tuned specialist across 29 cities. Per-city MASE difference (specialist – foundation model; positive favours the foundation model), sorted within panel; marker fill distinguishes data-rich from data-scarce cities ($n = 29$ cities per panel). **(a)** PM_{2.5} under the deployable causal covariates of Table 1: the foundation model is better in 21 of 29 cities (sign test $P = 0.024$), but no per-city difference is FDR-significant and the Wilcoxon test does not reach significance ($P = 0.084$), so there is no significant difference overall. **(b)** 2 m temperature at face value favours the specialist; this panel uses perfect-foresight covariates (the `future_covariates` convention; sign test $P < 0.001$), and Fig. 3 shows the lead disappears under causal covariates.

On PM_{2.5}, both the specialist and the foundation model cleared the seasonal-naïve floor (naïve MASE 1.026 ± 0.530 ; 13 FDR-significant Chronos-Bolt wins over naïve). The NAS-searched model and the covariate-residual Chronos variant trailed (0.734 ± 0.352 and 0.797 ± 0.481), and a Friedman test across all five tiers rejected tier equivalence overall ($P < 0.001$ in both configurations). The tie is therefore specific to the specialist-versus-foundation-model comparison; the leaderboard as a whole is not flat.

The foundation model's advantage showed no detectable correlation with how much local history a city had: in the deployable causal configuration, the correlation between usable hours and the specialist-minus-FM MASE gap was $r = -0.031$ ($P = 0.873$, bootstrap 95% CI $[-0.287, 0.251]$), and the perfect-foresight panel is likewise flat ($r = 0.075$, $P = 0.698$). Zero-shot competitiveness is therefore not a scarce-data phenomenon that longer records would erase. This is a cross-sectional check across heterogeneous cities and is read descriptively; usable hours is a coarse proxy, since the zero-shot model reads a fixed four-week context while trained tiers can exploit the full history.

Table 1. Panel accuracy by domain, covariate configuration, and tier. MASE, MAE, and RMSE (mean $\pm$ sd across 29 cities; NAS-GRU collapsed to per-city seed-means first; MAE/RMSE in target units, $\mu\text{g m}^{-3}$ for PM2.5 and $^{\circ}\text{C}$ for temperature), Friedman–Nemenyi average rank across the five tiers (lower is better; critical difference 1.13 at $\alpha = 0.05$, $k = 5$, $n = 29$), and FDR-corrected Diebold–Mariano significant-win counts against zero-shot Chronos-Bolt (tier wins / foundation-model wins). Each domain carries two blocks: the deployable *causal* configuration (weather covariates at their last value known at the forecast origin; the primary result, as in Figs. 2a and 3) and the *perfect-foresight* upper bound (covariates at the forecast target time; per-city detail in Supplementary Table S13). The univariate zero-shot and naïve tiers use no weather covariates, so their accuracy is identical in both blocks; their rank still shifts because competitors move. Under causal covariates no tier records an FDR-significant win over the zero-shot model in either domain, and the specialist–foundation-model rank gap is inside the critical difference in both domains (PM2.5: 2.41 vs 1.72; temperature: 1.72 vs 1.93); the perfect-foresight temperature block shows the 6-win specialist upper bound that the causal ablation removes.

Domain	Covariates	Tier	MASE (mean $\pm$ sd)	MAE (mean $\pm$ sd)	RMSE (mean $\pm$ sd)	Avg. rank	FDR wins (tier / FM)
PM _{2.5}	Causal (deployable)	Chronos-Bolt (zero-shot)	0.662 $\pm$ 0.368	10.553 $\pm$ 13.524	15.093 $\pm$ 19.273	1.72	—
		Chronos-Bolt + covariates	0.797 $\pm$ 0.481	13.500 $\pm$ 19.643	18.425 $\pm$ 26.624	3.34	0 / 5
		LightGBM (specialist)	0.692 $\pm$ 0.374	10.902 $\pm$ 13.617	15.245 $\pm$ 19.246	2.41	0 / 0
		NAS-GRU	0.734 $\pm$ 0.352	11.426 $\pm$ 14.931	15.592 $\pm$ 20.791	2.90	0 / 2
		Seasonal-naïve	1.026 $\pm$ 0.530	16.505 $\pm$ 21.235	22.274 $\pm$ 28.983	4.62	0 / 13
	Perfect foresight	Chronos-Bolt (zero-shot)	0.662 $\pm$ 0.368	10.553 $\pm$ 13.524	15.093 $\pm$ 19.273	1.83	—
		Chronos-Bolt + covariates	0.730 $\pm$ 0.376	11.745 $\pm$ 15.266	15.959 $\pm$ 20.478	3.14	1 / 2
		LightGBM (specialist)	0.662 $\pm$ 0.371	10.845 $\pm$ 14.060	14.810 $\pm$ 18.895	2.17	0 / 0
		NAS-GRU	0.734 $\pm$ 0.352	11.426 $\pm$ 14.931	15.592 $\pm$ 20.791	3.07	0 / 2
		Seasonal-naïve	1.026 $\pm$ 0.530	16.505 $\pm$ 21.235	22.274 $\pm$ 28.983	4.79	0 / 13
Temperature	Causal (deployable)	Chronos-Bolt (zero-shot)	0.792 $\pm$ 0.282	1.303 $\pm$ 0.717	1.819 $\pm$ 1.040	1.93	—
		Chronos-Bolt + covariates	2.614 $\pm$ 1.044	3.923 $\pm$ 1.821	4.919 $\pm$ 2.217	4.72	0 / 27
		LightGBM (specialist)	0.745 $\pm$ 0.223	1.242 $\pm$ 0.701	1.629 $\pm$ 0.903	1.72	0 / 1
		NAS-GRU	0.999 $\pm$ 0.481	1.626 $\pm$ 1.093	2.070 $\pm$ 1.261	2.59	0 / 6
		Seasonal-naïve	1.686 $\pm$ 1.050	2.716 $\pm$ 1.780	3.300 $\pm$ 2.085	4.03	0 / 16
	Perfect foresight	Chronos-Bolt (zero-shot)	0.792 $\pm$ 0.282	1.303 $\pm$ 0.717	1.819 $\pm$ 1.040	2.90	—
		Chronos-Bolt + covariates	0.718 $\pm$ 0.278	1.228 $\pm$ 0.795	1.559 $\pm$ 1.012	2.41	1 / 2
		LightGBM (specialist)	0.533 $\pm$ 0.208	0.888 $\pm$ 0.550	1.153 $\pm$ 0.730	1.28	6 / 0
		NAS-GRU	0.999 $\pm$ 0.481	1.626 $\pm$ 1.093	2.070 $\pm$ 1.261	3.59	0 / 6
		Seasonal-naïve	1.686 $\pm$ 1.050	2.716 $\pm$ 1.780	3.300 $\pm$ 2.085	4.83	0 / 16

The specialist's temperature edge depends heavily on covariate timing

At face value the temperature domain looked like a specialist win. With perfect-foresight covariates (reported as an upper bound in Supplementary Table S13), LightGBM reached panel-mean MASE 0.533 ± 0.208 versus 0.792 ± 0.282 for zero-shot Chronos-Bolt, with 6 FDR-significant specialist wins and none in the other direction (sign and Wilcoxon $P < 0.001$) (Fig. 2b). That comparison, however, is a perfect-foresight one: it hands every covariate-using tier the weather at the forecast *target* time (Methods). For PM2.5 the assumption adds little, because weather is a weak 24-h predictor of pollution. For temperature the covariates (humidity, pressure, radiation, dew point) are near-deterministic physical drivers of the target, so the assumption carries substantial weight.

We therefore re-ran the specialist, the strongest covariate user and the tier carrying the domain claim, under causal covariates. The restriction removed the temperature lead (Fig. 3). Panel-mean MASE rose from 0.533 to 0.745, the per-city win rate against Chronos-Bolt fell from 26/29 (Wilcoxon $P = 1.4 \times 10^{-6}$) to 16/29 ($P = 0.29$, no longer significant), and the foresight assumption alone was worth +0.212 MASE ($P = 2.6 \times 10^{-8}$). In the causal-covariate panel that Table 1 now reports as the main result, the specialist records zero FDR-significant DM wins over the foundation model, and the foundation model records one (0/1), a reversal of the 6/0 count seen under perfect foresight. The same restriction hits the covariate-residual Chronos variant far harder in this domain, because its ridge component extrapolates stale covariates across the horizon. Its panel-mean MASE degrades to 2.614 ± 1.044 , worse than the seasonal-naïve floor (Table 1). The same ablation left PM2.5 unmoved (0.662 perfect-foresight vs 0.692 causal, both statistically indistinguishable from Chronos-Bolt at 0.662; $P = 0.33$ and $P = 0.08$). Held to the stricter equivalence standard, the two domains differ only in which direction the residual uncertainty leans. On PM2.5, formal equivalence at the 0.05-MASE margin is established in the perfect-foresight configuration (above); under causal covariates it is not, and the interval leaves room only on the specialist's unfavourable side (90% CI $[-0.012, 0.071]$). On temperature under causal covariates equivalence is likewise not established, but the interval leans the specialist's way: the paired difference favours it by a small, non-significant margin (0.745 vs 0.792; 90% CI $[-0.114, 0.018]$). The corrected cross-domain finding is therefore not a domain flip but a graded one. Under realistic covariate availability, the small zero-shot foundation model shows no statistically significant accuracy deficit to the tuned specialist in either domain (formal equivalence is established on PM2.5 when the specialist is granted perfect-foresight covariates), and much of the specialist's large apparent weather edge is attributable to assuming a perfect forecast of its own inputs. How much is quantified in the next section, which replaces that assumption with a measured one rather than removing it. Perfect-foresight numbers are retained in all tables as a clearly-labelled upper bound on specialist performance. As in PM2.5, the temperature-domain FM advantage showed no detectable linear association with data volume in the causal configuration (Pearson $r = 0.138$, $P = 0.477$, 95% CI $[-0.220, 0.452]$). With only 29 cities this is an absence of detectable association, not evidence of true independence.

Realistic forecast covariates place the temperature edge between the two extremes

The causal and perfect-foresight regimes bracket deployment rather than describe it. To locate the deployable point inside that bracket we measured the true error of archived 24-hour weather forecasts for every covariate the specialist consumes and re-ran the panel with covariates degraded to that measured quality, sweeping a scale factor α over the covariate-quality axis (Methods; Fig. 6).

On temperature the specialist's advantage over the zero-shot foundation model falls monotonically as covariate quality degrades, and the realistic point sits well inside the bracket rather than at either end: with perfect foresight ($\alpha = 0$) the specialist reaches 0.533 MASE, a lead of +0.260 over the zero-shot model; with covariates degraded to the measured real-forecast error level ($\alpha = 1$) it reaches 0.638, a lead of +0.154, or 59% of the perfect-foresight lead; and with last-known covariates it reaches 0.745, a lead of +0.048, or 18%.

A specialist supplied with covariates of genuine forecast quality therefore retains about three-fifths of the advantage that perfect foresight confers, against under one-fifth for the last-known floor. The result is not an artifact of a single noise draw: across replicate injections the panel mean varies by 0.009–0.018 MASE, roughly a sixth of the 0.106 MASE shift from perfect foresight to realistic covariates. As a check that the degradation reached only the intended path, the covariate-free tiers are bitwise identical at every α .

Two qualifications belong with this result. First, matching the *magnitude* of last-known error does not reproduce its *effect*: the panel does not reach the last-known regime's own 0.745 MASE until $\alpha \approx 3.4$ by extrapolation, more than twice the $\alpha = 1.54$ that matching error *variance* alone would predict, because holding a covariate flat removes information a matched-variance noise process does not (Methods). We therefore report the last-known figure directly rather than reading it off the α axis. Second, the width of the bracket depends on how much information the covariates carry about the target: on PM2.5 the perfect-foresight and last-known specialists differ by ~ 0.03 MASE to begin with, so there is almost no bracket to locate a point inside, which is the same domain asymmetry reported above.

The practical reading is that the future-covariates convention distorts a cross-domain comparison substantially but does not invent the specialist's temperature advantage. A deployment with a trustworthy weather feed should expect to keep most of that

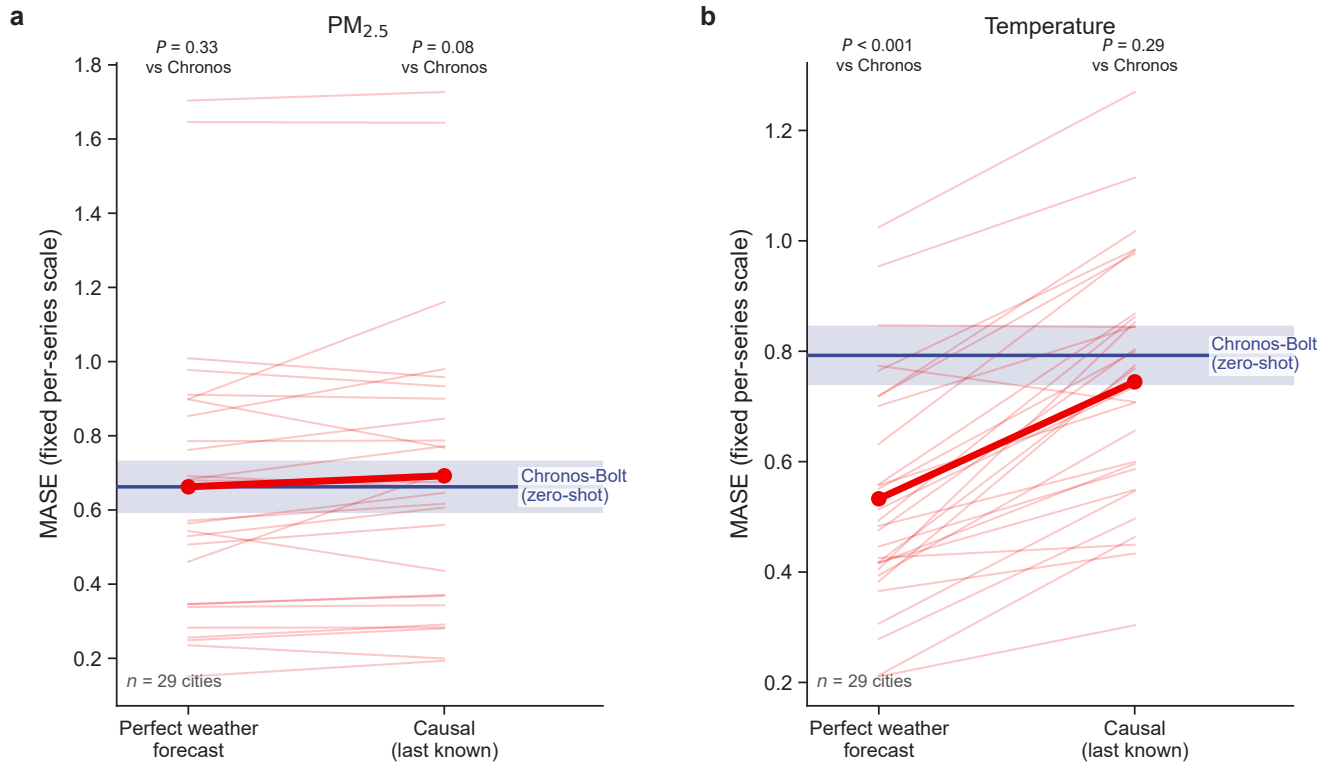


Figure 3. The specialist’s temperature edge depends heavily on covariate timing. Per-city LightGBM MASE with perfect-foresight covariates (left ticks) versus causal, last-known covariates (right ticks), $n = 29$ cities per panel; bold line = panel mean; horizontal band = zero-shot Chronos-Bolt panel mean $\pm$ s.e.m. across cities. On-panel P -values are paired Wilcoxon signed-rank tests of LightGBM against Chronos-Bolt. **(a)** $PM_{2.5}$: foresight is worth ~ 0 (0.662 vs 0.692; both statistically indistinguishable from the foundation model). **(b)** Temperature: removing foresight moves the specialist from 0.533 (26/29 city wins, $P = 1.4 \times 10^{-6}$) to 0.745 (16/29, $P = 0.29$, not significant) against the foundation model at 0.792.

advantage; one without should expect to lose it.

Beijing multi-station depth check

One sensor per city trades depth for breadth, so we added a 12-station within-city check⁴⁶. Zero-shot Chronos-Bolt beat the LightGBM specialist at 12 of 12 stations, and beat both the specialist and the seasonal-naïve floor at all 12 (Chronos-Bolt MASE range 0.153–0.297; LightGBM 0.292–0.458; naïve 1.03–1.24) (Fig. 4). The panel-level tie is therefore not an averaging artifact over heterogeneous stations.

The evaluated transfer recipe does not outperform zero-shot in the data-scarce case

The crux question inherited from our earlier conference study²⁴ is whether a NAS-discovered model *pretrained on data-rich cities and fine-tuned on a scarce city* still beats a foundation model that has never seen the target city. It does not. Across the 15 scarce-tier cities at fine-tune budgets of $\{0, 1, 10, 100\}$ % of local history (Methods), zero-shot Chronos-Bolt held the best mean MASE at every budget: 0.843 versus 0.899/0.915/0.888/0.876 for transferred NAS-GRU and 0.941/0.944/0.858 for a LightGBM refit on the same budget (Table 2). The transfer recipe never overtakes zero-shot: after Holm correction across fractions, the city-level paired Wilcoxon tests find no fraction at which transfer significantly beats zero-shot. The paired difference in fact favours the zero-shot model at every budget, nominally significantly at the 0% and 1% budgets (bootstrap 95% CI on the difference excludes zero) and non-significantly at 10% and 100%. Equivalence is not established at any budget (all TOST 90% CIs exceed the 0.05-MASE margin; Supplementary Fig. S2). We are therefore not claiming that the two are interchangeable, only that the previously published transfer recipe does not recover a data-scarce advantage over simply running the foundation model. Strategy choice in the scarce regime therefore turns on cost and operational simplicity, and there the zero-shot model wins (next subsection).

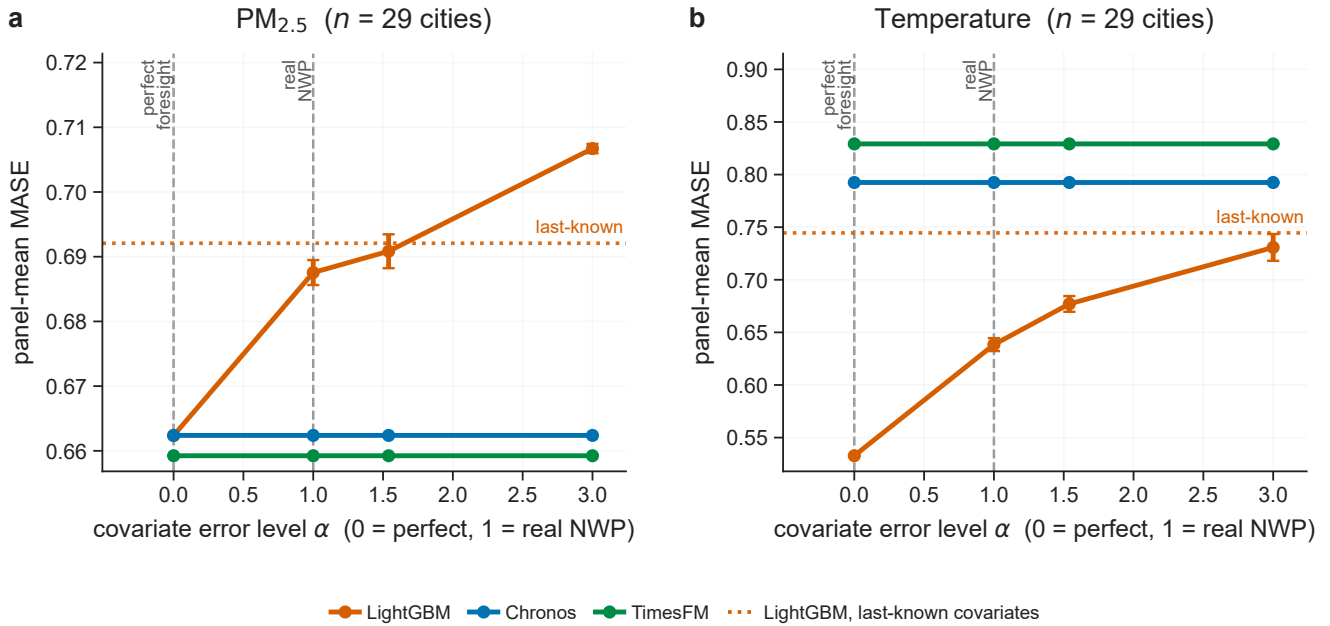


Figure 4. The specialist’s advantage as a function of covariate quality. Panel-mean MASE against α , the multiplier on measured 24-hour numerical-weather-prediction error injected into the specialist’s future covariates (Methods): $\alpha = 0$ reproduces the perfect-foresight regime exactly and $\alpha = 1$ is the measured real-forecast error level, calibrated per variable against the Open-Meteo previous-model-runs archive. (a) PM_{2.5}, where the whole covariate-quality axis spans only 0.044 MASE (0.662 to 0.707), so the specialist never overtakes either foundation model once its covariates are imperfect. Note the very different vertical scales: the PM_{2.5} panel is magnified roughly fivefold relative to the temperature panel, which makes a small absolute effect look steep. (b) Temperature, where the specialist retains roughly three-fifths of its perfect-foresight advantage at realistic covariate quality and under one-fifth at the last-known floor (dotted line). Error bars span replicate noise injections. The covariate-free foundation-model tiers are flat in α by construction, and their invariance doubles as a check that the degradation reached only the intended path. Only sweep cells covering the full 29-city panel are plotted.

Energy, cost, and the deployment decision

Under our backtesting-and-retraining protocol, the measured end-to-end energy reverses the expectation that the foundation model is the expensive option (Table 3). Because the specialist retrains one model per lead time per fold while the zero-shot model only runs inference, LightGBM consumed 9.5–15.1 kJ per 1,000 forecasts across the three replication cities, against 1.0–1.2 kJ for zero-shot Chronos-Bolt on a consumer GPU (one repeatability-flagged Nairobi cell ranged up to 5.4 kJ). That is roughly an order of magnitude in energy, and about $8\times$ in cost (6.9×10^{-4} versus 8.2×10^{-5} USD per 1,000 forecasts at 0.15 USD/kWh and a power-usage-effectiveness (PUE) factor of 1.4). NAS-GRU, which must be trained locally, sits with the specialist (7.3–16.8 kJ/1k). Cells that failed the repeatability gate are shown as ranges rather than point estimates throughout (4 of 15; Methods).

The direction survives a substantially larger foundation model. Measured on the Beijing depth city under the same protocol, TimesFM consumed 1.15 kJ per 1,000 forecasts against 0.53 kJ for Chronos-Bolt, a ratio close to their $4.8\times$ difference in parameter count, but still $2.8\times$ less than the 3.22 kJ the retrained specialist required on the same series. Model size therefore moves the foundation-model tier’s energy cost without moving the comparison it is part of, because the specialist’s cost is dominated by retraining rather than by inference. We report this as a single-city check rather than a panel result, since only Chronos-Bolt was measured across the three replication cities.

That comparison reflects a frequent-retraining deployment, not a train-once one, and the distinction matters. Separating one-time training energy from per-forecast inference energy shows the measured gap is almost entirely a training gap. At inference alone, the specialist’s 24 small per-horizon tree models are cheaper per forecast than a single Chronos-Bolt forward pass, in all three replication cities on CPU and in the two cities with a valid GPU estimate (the Beijing GPU inference measurement is a sub-kilojoule artifact; Supplementary Table S12). A specialist that is trained once and then reused therefore crosses over to being the more energy-efficient option after roughly 2,800–4,400 forecasts on CPU (≈ 120 –180 days of hourly 24-h-ahead forecasting for one series) and 250–2,100 on GPU. The order-of-magnitude energy advantage of the zero-shot model is thus specific to deployments that retrain frequently (more often than every few months); it is not a universal property.

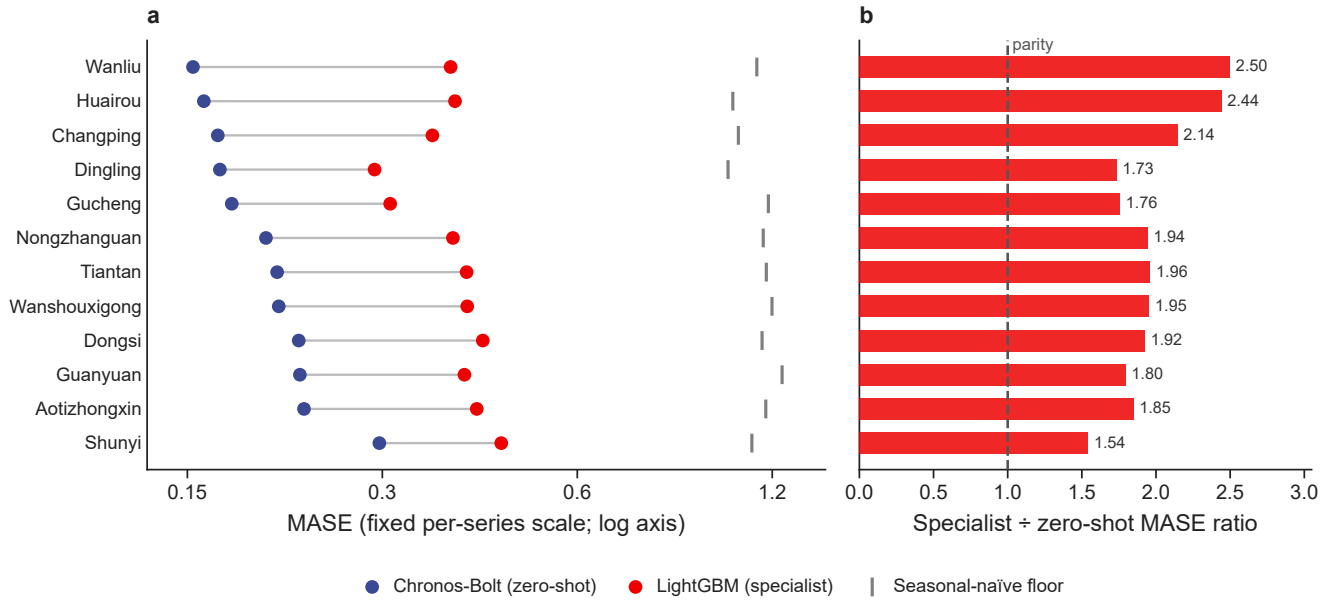


Figure 5. Beijing multi-station depth check. (a) Per-station MASE at the 12 UCI Beijing multi-site stations (six-fold backtest, after the pre-specified value-sanitization step; Methods). Zero-shot Chronos-Bolt (0.153–0.297) beats the LightGBM specialist (0.292–0.458) at 12/12 stations; the seasonal-naïve floor spans 1.03–1.24. (b) Specialist-to-foundation-model MASE ratio per station (same station order); the dashed line marks parity. The advantage is uniform across the metropolitan network rather than driven by a subset of stations, so the panel-level tie is not an averaging artifact.

Table 2. Transfer learning versus zero-shot across fine-tune data budgets (E4). Panel results over the 15 data-scarce cities: NAS-GRU pretrained on the 14 data-rich cities and fine-tuned at nominal budgets of {0, 1, 10, 100}% of local history (5 seeds, per-city seed-means first), and a LightGBM specialist refit on the same nominal budget (its window floors at 202 h, so no 0% row exists; the asymmetric data floors at the 1% budget are described in Methods). Columns: mean actual training hours behind each nominal budget, panel MASE (mean $\pm$ sd across cities), paired panel-mean difference versus zero-shot Chronos-Bolt (positive means zero-shot is better), per-city win counts (strategy / foundation model), Holm-corrected paired Wilcoxon P across budgets, and the TOST equivalence interval on the paired difference for the pre-specified transfer comparisons (computed for the confirmatory NAS-GRU family). Two quantities in this table share the value 0.05 but are unrelated: the equivalence margin $\delta = 0.05$ is measured in MASE units, whereas the significance level is $\alpha = 0.05$. The reported interval is a 90% one because TOST tests against the $1 - 2\alpha$ interval, not the 95% interval; equivalence at $\delta = 0.05$ is not established at any budget. Zero-shot Chronos-Bolt holds the best panel mean at every budget, and no transfer budget significantly beats it (all Holm $P \geq 0.19$). The LightGBM refits are significantly worse at the two smallest budgets (Holm $P = 0.009$ at the 1% budget and 0.006 at the 10% budget), but not at the 100% budget.

Strategy	Budget (%)	Hours (mean)	MASE (mean $\pm$ sd)	Δ vs. zero-shot (panel mean)	Wins (strat. / FM)	Holm P (Wilcoxon)	TOST 90% CI ($\delta = 0.05$)
Chronos-Bolt (zero-shot)	—	—	0.843 $\pm$ 0.354	—	—	—	—
NAS-GRU (transfer + fine-tune)	0	0	0.899 $\pm$ 0.340	+0.056	4 / 11	0.192	[+0.005, +0.107]
	1	66	0.915 $\pm$ 0.341	+0.072	5 / 10	0.250	[+0.015, +0.129]
	10	599	0.888 $\pm$ 0.350	+0.045	7 / 8	0.421	[-0.012, +0.102]
	100	5,995	0.876 $\pm$ 0.383	+0.033	4 / 11	0.338	[-0.009, +0.075]
LightGBM (refit on budget)	1	202	0.941 $\pm$ 0.311	+0.098	1 / 14	0.009	—
	10	599	0.944 $\pm$ 0.383	+0.100	1 / 14	0.006	—
	100	5,995	0.858 $\pm$ 0.382	+0.015	6 / 9	0.421	—

The electricity cost of either tier is below 10^{-3} USD per 1,000 forecasts, so in absolute monetary terms the difference is small next to hardware, memory, and engineering costs. The zero-shot model’s durable advantages are avoiding the training step altogether and its operational simplicity, not the electricity bill (Discussion).

Table 3. Measured energy, cost, and model size per tier and replication city. codecarbon-measured joules per 1,000 forecasts for every tier in each of the three replication cities (mean $\pm$ sd over five repetitions; min–max range shown where sd/mean $>$ 0.20), per-city cost at 0.15 USD/kWh with PUE 1.4 (the decision rule’s central assumptions), the per-city CPU train/inference decomposition (one-time training energy per fit, per-forecast inference energy, and the crossover, meaning the forecasts a once-trained specialist serves before its amortized energy per forecast falls below the foundation model’s; measured only for the two tiers in the amortization protocol, GPU rows in Supplementary Table S12), and parameter count as a hardware-independent size proxy (LightGBM’s tree ensemble has no directly comparable unit and is not reported). Measured J/1k figures include per-fold training for trained tiers; the zero-shot tiers run inference only. The covariate-residual variant measures below plain zero-shot because tiers run sequentially within each measured workload and the zero-shot tier, which runs first, absorbs the one-time GPU initialization and model-load cost. Per-forecast latency and peak memory were not logged for the canonical runs.

Tier	City	Measured (J/1k)	Cost (USD/1k)	Train (J/fit, CPU)	Inference (J/forecast, CPU)	Crossover (forecasts, CPU)	Params
Chronos-Bolt (zero-shot)	Beijing	1,229 $\pm$ 241	7.2×10^{-5}	—	0.112	—	47.72M
	Seoul	1,025 $\pm$ 141	6.0×10^{-5}	—	0.110	—	
	Nairobi	1,056–5,417	1.1×10^{-4}	—	0.111	—	
Chronos-Bolt + covariates	Beijing	763 $\pm$ 102	4.5×10^{-5}	—	—	—	47.72M
	Seoul	781 $\pm$ 94	4.6×10^{-5}	—	—	—	
	Nairobi	842 $\pm$ 109	4.9×10^{-5}	—	—	—	
LightGBM (specialist)	Beijing	15,061 $\pm$ 2,374	8.8×10^{-4}	317	0.033	4,041	—
	Seoul	10,832 $\pm$ 406	6.3×10^{-4}	246	0.022	2,778	
	Nairobi	9,461 $\pm$ 836	5.5×10^{-4}	203	0.065	4,416	
NAS-GRU	Beijing	9,367–16,788	7.7×10^{-4}	—	—	—	157,080
	Seoul	7,496 $\pm$ 440	4.4×10^{-4}	—	—	—	
	Nairobi	7,298 $\pm$ 105	4.3×10^{-4}	—	—	—	
Seasonal-naïve	Beijing	2 $\pm$ 0	1.2×10^{-7}	—	—	—	—
	Seoul	1–36	5.3×10^{-7}	—	—	—	
	Nairobi	1–40	9.3×10^{-7}	—	—	—	

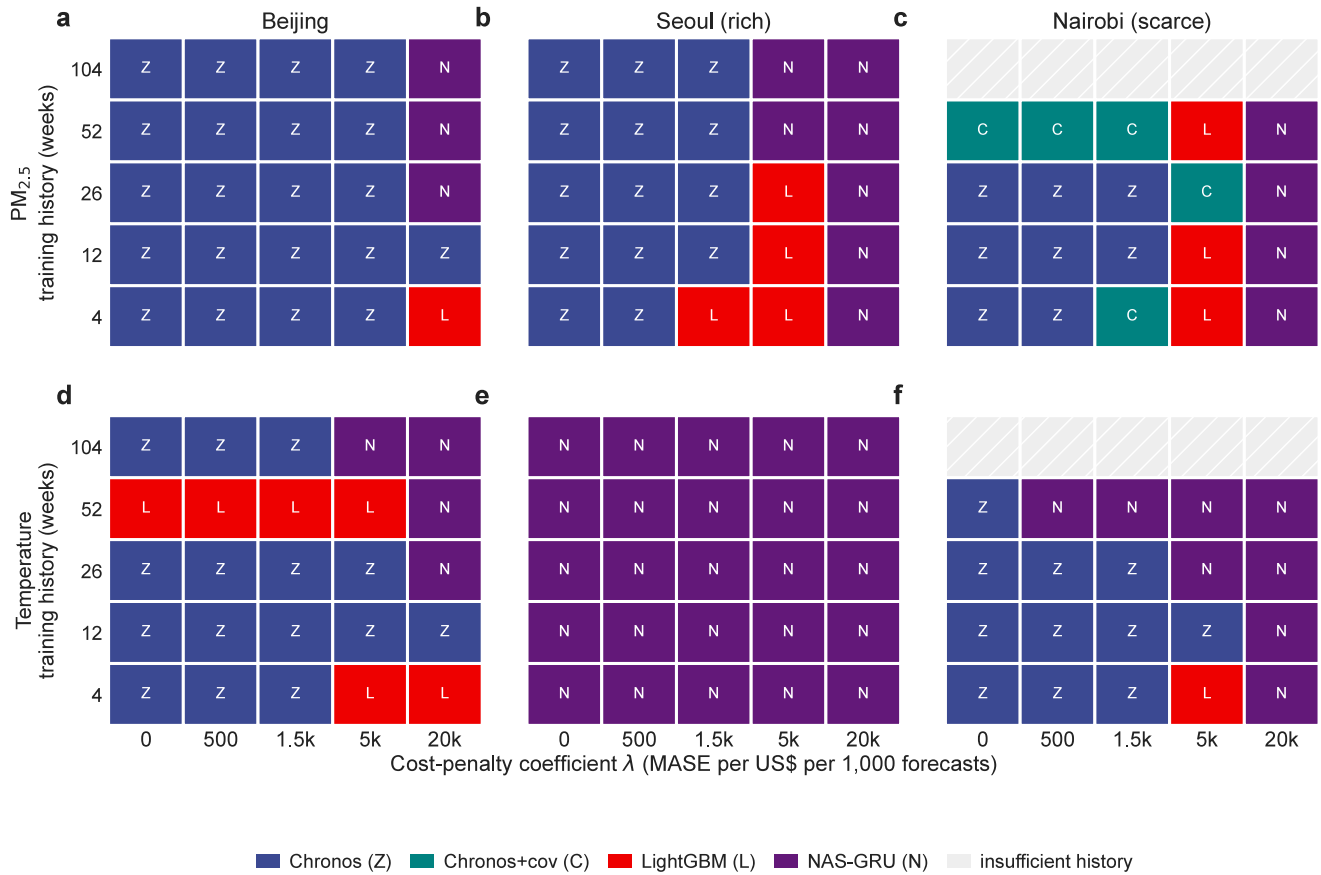


Figure 6. Cost-adjusted deployment winner maps (causal covariates). Winning tier under the objective $\text{MASE} + \lambda \cdot \text{USD}/1,000$ forecasts, per training-history regime (rows: weeks of history) and cost-penalty coefficient λ (columns), for Beijing, Seoul, and Nairobi in both domains (a–f), using the deployable causal (last-known) covariates. PM2.5 maps are dominated by the Chronos family (21/25, 14/25, and 13/20 cells). Under causal covariates the temperature maps no longer favour the specialist: the Chronos family and NAS-GRU take most cells (e.g. Beijing 15/25 Chronos vs 6/25 specialist; Seoul 25/25 NAS-GRU), a marked shift from the perfect-foresight version (Supplementary Fig. S1), consistent with the ablation (Fig. 3). Sweeping electricity price over 0.05–0.30 USD/kWh and PUE over 1.0–2.0 flips no cells at the central assumption and at most 40% at the most extreme corners (Supplementary Table S11). Cell letters duplicate the colour coding for grayscale readers.

Folding cost into the objective produces deployment winner maps across training-history regimes for one depth city (Beijing) plus one rich (Seoul) and one scarce (Nairobi) panel city in both domains (Fig. 5). Each cell selects the tier minimizing $\text{MASE} + \lambda \cdot \text{USD}/1k$ over the sweep of the cost-penalty coefficient λ (Methods): $\lambda = 0$ selects on accuracy alone, and larger λ penalizes cost more heavily. The main maps use the deployable causal covariates; a perfect-foresight version is a labelled upper bound (Supplementary Fig. S1). In PM2.5 the two configurations are nearly identical (weather is barely informative there) and the foundation-model family wins most of the causal map: 21/25 cells in Beijing, 14/25 in Seoul, and 13/20 in Nairobi to the Chronos family (8 zero-shot + 5 covariate variant). In temperature the configurations diverge sharply. Under perfect foresight the specialist takes many cells (11/25 Beijing, 12/25 Seoul, 18/20 Nairobi), but under causal covariates its share collapses: the Chronos family and NAS-GRU take most cells (Beijing 15/25 Chronos vs 6/25 specialist; Seoul 25/25 NAS-GRU; Nairobi 11/20 Chronos plus 8/20 NAS-GRU), consistent with the ablation. The maps are robust to the cost assumptions at the central setting and degrade gracefully away from it: across the full electricity-price $\times$ PUE sweep, no cells flip at the central assumption and at most 40% flip at the most extreme corners (Supplementary Table S11).

Uncertainty calibration

Split-conformal 95% intervals were near nominal for all tiers in both domains (Supplementary Table S4; per-city coverage and width in Supplementary Tables S6–S7). In the causal-covariate panel, pooled empirical coverage ranged 0.910–0.970 on PM2.5 and 0.900–0.968 on temperature; the low end is the weather rich-tier zero-shot pool at 0.90 against the 0.95 nominal. Under

these realistic covariates the specialist's temperature intervals (rich-tier mean width 9.6 °C-scale units) were only marginally narrower than zero-shot Chronos-Bolt's (10.7), and most of the sharpness the perfect-foresight configuration gave it (7.7, Supplementary Table S4) is lost. On PM2.5 the foundation model's intervals were in fact tighter than the specialist's (rich-tier 18.5 vs 21.0). The naïve floor achieved coverage only through much wider intervals (35.1 vs ~18 on rich-tier PM2.5).

Discussion

The efficiency calculus has changed

The premise of our earlier conference study was that foundation models are too computationally heavy for edge deployment, so data-scarce cities need searched architectures and transfer learning²⁴; the assumption was widely shared at the time and, there as elsewhere, was carried as a design constraint rather than measured. This study keeps the same evaluation philosophy and supplies the missing comparison. With a compact foundation model in the panel, the constraint no longer binds. Under realistic covariate availability, zero-shot Chronos-Bolt is statistically indistinguishable from the tuned specialist on PM2.5 (and equivalent by a formal TOST when the specialist is given perfect covariates), no longer significantly worse on temperature, and never significantly beaten by the transfer recipe at any fine-tune budget, where it leads at every one. Under a frequent-retraining protocol it also uses roughly an order of magnitude less energy, because it never trains.

Why should a model that has never seen a city forecast it about as well as a model trained there? A plausible reading is that the quantities a local specialist must estimate from a short, gappy record (the shape of the diurnal cycle, weekly structure, the persistence of pollution episodes) are the same generic regularities a foundation model has already absorbed from large, heterogeneous pretraining corpora²⁵, so a four-week context window suffices to localize them. This reading is consistent with our observation that the zero-shot advantage shows no detectable association with local record length. It also sharpens, rather than contradicts, recent skepticism about foundation-model leaderboards^{32,33}. In our panel the zero-shot model does not beat a well-tuned gradient-boosting specialist, which extends the long record of simple methods matching complex ones³⁴; but matching the specialist *without training* is precisely what changes the deployment decision. The earlier work's comparisons remain internally sound, and search remains the right tool when an application needs a bespoke model; what changed is that small TSFMs^{25,42} opened a second route to the same destination and moved the accuracy–cost frontier with it. Reducing model size by search and avoiding training altogether now reach the same accuracy band, and the second is operationally simpler. For an operator in a data-scarce city, the practical default shifts from training a small model to downloading one.

A covariate-timing pitfall that can produce spurious domain conclusions

A cross-domain claim can be an artifact of covariate timing alone. Had we used the standard `future_covariates` convention uncritically, this paper would have reported a clean domain flip (foundation model wins PM2.5, specialist wins temperature), with 26/29 significant city wins backing the specialist's temperature lead. The causal ablation shows that result depends substantially on granting the specialist a perfect forecast of its own near-deterministic drivers: with last-known covariates the specialist's lead is no longer statistically significant (16/29 cities, $P = 0.29$), shrinking from +0.259 to +0.047 MASE, though a paired equivalence test does not certify the two as equivalent in this domain. The asymmetry is domain-dependent (worth +0.212 MASE in temperature and ~0 in PM2.5) because covariate informativeness about the target differs by domain, which is precisely why the artifact is hard to detect in cross-domain comparisons: the same convention, applied uniformly, distorts one domain and not the other. Forecast-evaluation guides catalogue leakage and backtest pitfalls^{43,44}; we add covariate foresight to that list and recommend that benchmarks using future covariates report a causal-covariate ablation alongside, or at minimum report both configurations to bracket where reality falls between them. An operator with access to NWP-grade covariate forecasts sits between our causal floor and perfect-foresight ceiling⁶, and we locate that operating point rather than leaving it bracketed. Calibrating the covariates against measured 24-hour forecast error places the specialist at 0.638 MASE, a lead of +0.154 over the zero-shot model: about three-fifths of the perfect-foresight advantage, against under a fifth for the last-known floor. Two consequences follow. The convention's distortion is real and large enough to reverse a cross-domain reading, so a causal ablation remains necessary; but the distortion does not dissolve the specialist's temperature advantage, and a deployment with a trustworthy weather feed should expect to keep most of it. Reporting only the two extremes would have overstated the correction in one direction exactly as the uncritical convention overstates it in the other.

Cost-aware deployment guidance

The winner maps translate these statistics into an operational decision that turns on three measured quantities: available local history, covariate-forecast quality, and refresh cadence.

Local history matters least. The zero-shot model already takes most cells at the shortest regime mapped, four weeks of training history, and the size of its advantage is unrelated to record length across the panel, so an operator choosing on accuracy alone at realistic covariate availability has little reason to train.

Covariate-forecast quality matters most, and it is the condition an operator is most likely to misjudge. Granted perfect-foresight weather the specialist takes many temperature cells, but restricted to what is knowable at the forecast origin that share collapses. Most operators sit between those cases, and the width of the bracket, +0.212 MASE in temperature and approximately zero in PM2.5, measures the value of an NWP-grade covariate feed⁶ against the cost of training a specialist to consume it.

Refresh cadence, not the electricity price, decides which option is cheaper. Under frequent retraining the zero-shot model uses roughly an order of magnitude less energy and about eight times less money because it never trains, but training amortizes: a specialist trained once crosses over after roughly 2,800–4,400 forecasts on CPU, about 120–180 days of hourly 24-hour-ahead service for one series. Because price and PUE act on the objective as linear rescalings of λ , a different tariff slides an operator along that axis rather than reordering the options, with no cell flips at the central assumption and at most 40% at the most extreme corners.

Two limits bound this advice. The electricity cost of either tier is below 10^{-3} USD per 1,000 forecasts, so the energy argument concerns scale and operational simplicity rather than the utility bill; for a single city, hardware and engineering time dominate. The E4 result also removes the middle option, since the transfer recipe neither beats zero-shot on accuracy in the scarce regime nor is demonstrably equivalent to it. This is the deployment analogue of arguments that energy belongs in the evaluation objective^{35,36,47}. Supplementary Table S16 sets out the procedure for re-deriving these maps for a new city.

Limitations

Seven limitations bound our claims. First, energy and hardware. CPU-side consumption is not metered: the study machine runs Windows, where RAPL is unavailable, so codecarbon models CPU draw from the processor’s 65 W TDP entry scaled by process utilisation, while GPU energy is read from NVML hardware counters. Every CPU-side energy figure should therefore be read as a comparative, directional quantity, valid for ranking tiers on one machine under one protocol but not as an absolute or cross-processor measurement. The benchmarked hardware is a consumer GPU and a CPU-only configuration rather than a physical embedded device, so the “edge/CPU-class” figures are indicative, not Raspberry-Pi- or Jetson-class measurements; per-forecast latency and peak memory were not logged, so deployment-critical figures beyond energy are reported only as parameter count (Table 3). Because the measured comparison includes per-fold training for trained tiers, the headline energy gap reflects frequent-retraining deployment; our amortization analysis reports the once-trained crossover separately. Second, contamination. Chronos-Bolt is pretrained on large public time-series collections whose exact composition we cannot fully audit, so we cannot rule out that the UCI Beijing series or ERA5/OpenAQ-derived data entered its corpus. To guard against this we re-evaluated on strictly post-2024-10 city windows (10 cities with sufficient coverage, a subset that by construction skews toward well-instrumented cities, since it requires a recent continuous record), which postdate the model’s pretraining corpus⁴² and cannot have been seen: the zero-shot model remained competitive with the specialist there (mean MASE 0.415 vs 0.395, Chronos-Bolt better in 6 of 10 cities, both far below the seasonal-naïve floor; Supplementary Table S14), so the PM2.5 result is not an artifact of pretraining exposure. We nonetheless treat the public Beijing 12-station set as corroborating rather than primary evidence. Third, coverage. The panel uses one sensor per city, so within-city spatial generalization rests on the Beijing 12-station check alone. Fourth, model scope. The foundation-model tier is Chronos-Bolt-small; larger TSFMs or future revisions may shift the accuracy–cost frontier, and our conclusions are explicitly frontier-relative, not model-family-absolute^{32,33}. Fifth, task and comparator scope. We test hourly, 24–48 h direct multi-horizon point forecasting, with each tier at its deployed context length (Chronos-Bolt four weeks, NAS-GRU 24 h). A context-length-matched comparison (168 h and 672 h GRU lookbacks) is a natural sensitivity check; our attempts to run it exceeded the memory of the study GPU (8 GB), so we report the comparison at native contexts and flag context length as a factor we did not isolate on this hardware. Longer horizons, probabilistic scoring rules, and multivariate targets are likewise untested here. Sixth, statistics. Split-conformal exchangeability holds only approximately under temporal dependence^{48,49}, and with 15 scarce cities the E4 tests have limited power. Even where we establish equivalence (PM2.5 under perfect-foresight covariates), it is at a 0.05-MASE margin, and where we do not (causal PM2.5, temperature, E4) “no significant difference” remains a parity statement, not proof of interchangeability. Seventh, period heterogeneity. Each city contributes whichever window its source record supports, so the panel is not contemporaneous, and ten of the 29 cities (34% of usable hours, seven scarce-tier) overlap the 2020–21 pandemic period. Because every comparison is within-city and between-tier against a per-series scale, a local regime shift rescales the problem for all tiers at once rather than favouring one, and the post-cutoff check above, which lies entirely outside that period, reproduces the result. A stratified re-analysis adds to this: both domains’ panel conclusions hold in the pandemic-exposed, unexposed, and heavily exposed strata alike, with no stratum significant after correcting the ten subgroup tests for multiplicity. Two caveats apply: the exposure correlates with the rich/scarce contrast rather than spreading evenly across it, and the smaller strata are less powerful, so a null within one is not positive evidence of equivalence (Supplementary Information).

367 Outlook

368 Two extensions follow directly. Light fine-tuning of small TSFMs on pooled regional data (the transfer recipe applied to the
369 foundation model rather than the searched model) would test whether the two strategies' strengths combine. The decision-rule
370 framework also generalizes: any deployment context can re-derive winner maps by inserting local electricity prices, hardware,
371 retraining frequency, and its own cost-penalty coefficient, since all code, analysis plans, and per-claim provenance ledgers are
372 released. As monitoring networks in low- and middle-income countries grow^{4,5}, our results indicate that the marginal cost of a
373 competent forecast for a new city is now close to the cost of running a small pretrained model's inference loop, which bears
374 directly on how air-quality early-warning capacity can be scaled^{1,2}.

375 Methods

376 Study design

377 We compare three deployment strategies for hyper-local hourly forecasting in data-scarce cities: *search* (a NAS-discovered
378 tiny recurrent model), *specialize* (a gradient-boosting model with covariates), and *zero-shot foundation model*. With the
379 covariate-residual foundation-model variant and a seasonal-naïve floor, these three strategies are realized as the five model tiers
380 reported throughout. Both domains, urban PM2.5 (primary) and 2 m air temperature (secondary), use the same 29-city panel
381 and the same evaluation harness. The study extends our earlier conference paper²⁴, from which it inherits the NAS-discovered
382 architecture, the weather domain, and the edge/energy framing; the foundation-model tier, the cost-adjusted decision rule, the
383 air-quality domain, the cross-domain generality test, and the transfer-versus-zero-shot crux experiment are new.

384 Data sources

385 **PM2.5.** Hourly PM2.5 from the OpenAQ v3 API⁴, one sensor per city (sensor IDs in `cities_manifest.csv`). For each
386 candidate city we selected the single OpenAQ sensor with the longest gap-gated usable window (the deterministic selection
387 rule; ties broken by higher pre-interpolation coverage), so “reference sensor” here means the best-covered instrument available
388 for that city rather than a certified reference-grade monitor; OpenAQ aggregates heterogeneous instruments, and we do not
389 assume uniform calibration across cities. Retrieval was month-windowed, resumable, and checkpointed, with per-request
390 record-count verification (Supplementary Information). **Temperature and meteorological covariates.** The Open-Meteo
391 historical archive⁵⁰, a gap-free reanalysis product built on ERA5⁵¹: 2 m temperature as the target (weather domain) plus the
392 seven surface covariates used in our conference study: relative humidity, surface pressure, 10 m wind speed and direction,
393 precipitation, cloud cover, and shortwave radiation. In the temperature domain the covariate set excludes the target and contains
394 no variable mathematically derived from it. Because these are reanalysis fields rather than a single physical station, the
395 temperature target is a location-specific gridded series; we describe it as location-level rather than station-level, and note that its
396 “data scarcity” is imposed by clipping to each city's PM2.5 window (Section on panel construction) rather than being intrinsic to
397 the gap-free weather record. **Beijing depth check.** The UCI Beijing Multi-Site Air Quality dataset (12 stations, hourly, March
398 2013–February 2017, meteorology included)⁴⁶.

399 Data quality gating and preprocessing

400 PM2.5 readings ≤ 0 or $\geq 985 \mu\text{g m}^{-3}$ were treated as missing before any gating. Non-positive concentrations are physically
401 invalid for PM2.5 mass; 985 is a recurring sensor/API sentinel value rather than a genuine extreme episode. Both rules were
402 fixed before the campaign, and panel membership is insensitive to the exact ceiling (Supplementary Information). From each
403 sanitized series we extracted a usable window by splitting at gaps longer than 48 h, interpolating only gaps ≤ 6 h inside the
404 kept segment, and requiring a final window of $\geq 2,160$ h (90 days) with $\geq 60\%$ real (pre-interpolation) observations. All gate
405 parameters were fixed before the campaign, not tuned post hoc. Mumbai's sub-hourly records were resampled to the hourly
406 grid before gating. Three cities, London, Kathmandu, and Kolkata, failed those gates and were excluded (Supplementary
407 Information). The final panel is 29 cities: 14 data-rich and 15 data-scarce (Supplementary Table S2).

408 Because each city contributes whichever window its source record supports, the panel is not contemporaneous: usable
409 windows span 2016 to 2026, and ten cities overlap the 2020–21 pandemic period. We did not exclude that period, and we report
410 the exposure and its consequences for the panel comparisons in the Supplementary Information.

411 The rich/scarce tier was assigned a priori, when the candidate-city list was fixed and before any data were retrieved, by
412 the maturity of each city's regulatory air-quality monitoring infrastructure: data-rich cities are served by long-established
413 government monitoring programmes (North America, Europe, developed Asia-Pacific, and the established networks of Santiago,
414 Mexico City, and Bangkok), while data-scarce cities are those whose OpenAQ coverage comes mainly from recent, sparse, or
415 low-cost/donor-funded deployments (South and Southeast Asia, Africa, and the Andean cities). The label is deliberately an
416 infrastructure-context classification, not a function of realized record length (several scarce-tier cities end up with long usable
417 windows, e.g. Nairobi; Supplementary Table S2), so tier membership was never revised after the data were seen.

For the temperature domain, each city’s forecasting window was clipped to that same city’s PM2.5 usable window, so cross-domain comparisons isolate the domain effect rather than confounding it with longer weather histories. The temperature loader does not extrapolate beyond observed coverage: windows are clipped to real weather coverage and only gaps ≤ 6 h are interpolated; a window that would require extrapolation is excluded rather than filled. An earlier loader version forward-filled beyond coverage for four cities and produced an implausible MASE; the error was identified during internal review, and all affected artifacts were regenerated under the corrected loader (deviations log).

Forecasting task and evaluation protocol

Hourly cadence; primary horizon $h = 24$ with direct multi-horizon forecasting (one specialist model per lead time); $h = 48$ as a supplementary sensitivity. Rolling-origin backtesting⁴⁴ with six folds, all evaluated on the final common test window per series, so that different training-history regimes are compared on identical test data. Accuracy is reported as MASE⁵² with one fixed scale per series, namely the in-sample error of the seasonal-naïve ($m = 24$) forecast over the pre-test training history; the scale is shared across all regimes for that series so MASE remains comparable when training history varies. MAE, RMSE and normalized variants are recorded in every artifact.

Model tiers

Seasonal-naïve. 168-h persistence (same hour last week), the explicit floor tier. Note this weekly-persistence floor is a distinct model from the daily ($m = 24$) naïve whose in-sample error sets the fixed MASE denominator; the two serve different roles (a competitor tier versus a scale) and are not required to coincide. **LightGBM specialist.** One direct model per lead time⁹ with lagged target, calendar features, and meteorological covariates; retrained per fold; `random_state = 42`. (A pre-campaign audit found `subsample = 0.8` set without `subsample_freq`, making bagging inert; the configuration was corrected and is documented in the hyperparameter table, Supplementary Table S3.) **NAS-GRU.** The Green-NAS-A architecture (two stacked GRU layers, 128 units) discovered by NSGA-II multi-objective search in our conference study^{24,53}; trained with Adam (learning rate 10^{-3}), early stopping (patience 10, ≤ 50 epochs), 24-h lookback. Because training is stochastic, it was run at five seeds {42–46} and reported as mean $\pm$ sd; no best-seed selection. **Chronos-Bolt (zero-shot).** `amazon/chronos-bolt-small`^{25,42} with a four-week context window and no training or fine-tuning of any kind; the point forecast is the model’s predicted mean. **Chronos-Bolt + covariates.** A residual scheme: a ridge regression on calendar and weather covariates is fit over the training history, the foundation model forecasts the residual (its temporal strength), and the covariate model’s prediction for the horizon is added back. This tier consumes future covariates and is therefore subject to the same perfect-foresight/causal distinction as the specialist (below); in the causal configuration the weather covariates over the horizon are frozen at their last-known value at the forecast origin before the covariate model is evaluated.

Additional foundation models (generality check). To separate properties of zero-shot foundation models from properties of one architecture and one checkpoint, the whole panel is additionally run with two further models that vary those two explanations one at a time. `amazon/chronos-bolt-base` (205 M parameters) changes checkpoint scale by 4.3-fold while holding family, developer and pretraining corpus fixed; `google/timesfm-2.5-200m-pytorch` (231 M parameters, decoder-only) changes family, developer and pretraining corpus at a scale matched to Chronos-Bolt-base. The comparison therefore isolates a scale effect within a family from a family effect at matched scale. Everything else is held constant: the same folds, four-week context window, metric, and covariate-residual scheme for each model’s covariate-consuming variant. TimesFM 2.5 provides a native covariate interface, which is not used, so that the covariate pathway is identical across families and the comparison concerns the foundation model rather than two covariate implementations. None of the checkpoints is trained or fine-tuned. Chronos-Bolt-small remains the tier carried through Table 1, the energy measurements and the cost-adjusted decision rule; the other two enter as accuracy generality checks.

Covariate timing (perfect foresight vs causal). We distinguish two covariate-timing regimes and fix their roles explicitly. **Primary (deployable): causal covariates.** Each meteorological covariate enters at its last value known at the forecast origin (calendar features remain future-known, since a calendar is deterministic); this is the configuration a real operator can run, and it is the one reported in Table 1, Fig. 2a, Fig. 3, and Fig. 5. **Ceiling (upper bound): perfect-foresight covariates.** Every covariate-using tier receives the meteorological covariate at the forecast *target* time (the standard `future_covariates` benchmark convention), equivalent to assuming a perfect weather forecast over the horizon; these numbers are reported only as a clearly-labelled upper bound on covariate-using-tier performance (Supplementary Table S13, Supplementary Fig. S1). The causal ablation was logged prospectively in the deviations record before it was run, and the full 29-city panel was re-run in both domains for both covariate-consuming tiers: the LightGBM specialist (the strongest covariate user and the tier carrying the domain claim) and the covariate-residual Chronos variant. NAS-GRU consumes only the past context window and is causal by construction. The univariate zero-shot tier uses no covariates and is identical in both regimes. This primary/ceiling hierarchy is stated identically in every table and figure caption that involves a covariate-using tier.

Realistic forecast covariates. The two regimes above bracket deployment rather than describe it: an operator would normally take future covariates from a numerical weather prediction (NWP) service, which is neither perfect nor frozen at the

origin. We therefore added a third, calibrated regime. Using the Open-Meteo previous-model-runs archive we measured the true error of archived forecasts for every covariate the specialist consumes, expressed as a fraction of each variable’s own standard deviation so that error levels transfer across cities. Forecast error is then injected into the covariate series as a stationary AR(1) process matched to the measured error magnitude, bias and lag-1 autocorrelation, with physical bounds enforced afterwards (non-negativity, percentage ranges, and a circular wrap for wind direction). A scalar α scales the injected error, giving a continuous covariate-quality axis on which $\alpha = 0$ reproduces the perfect-foresight regime exactly and $\alpha = 1$ is the measured real-NWP error level. Because the weather covariates enter the design matrix only as the future-covariate block, degrading the covariate series is equivalent to replacing perfect foresight with a forecast of the stated quality, and it does so identically for every covariate-consuming tier.

Two properties of the calibration archive constrain this construction and are detailed in the Supplementary Information: its coverage is rich-tier-biased (15 of 29 cities), so the error model is calibrated on covered cities and applied to all 29, and its lead times (24–47 h) exceed our forecast horizon (1–24 h), so error is regressed on lead time rather than taken at face value. A third property constrains interpretation directly: matching the error *magnitude* of the last-known regime does not reproduce its effect. Equating error variance predicts that regime at $\alpha = 1.54$, yet the panel does not reach it even at $\alpha = 3$ (0.731 against 0.745), and linear extrapolation places the equivalent near $\alpha \approx 3.4$. Holding a covariate flat for 24 h removes the diurnal cycle, a coherent distortion, whereas AR(1) noise of the same variance blurs that cycle but preserves it, so persistence costs appreciably more than its variance implies. We therefore report the last-known result directly rather than reading it off the α axis, treating the variance-matched equivalent as a lower bound.

E4: transfer learning versus zero-shot (crux experiment)

NAS-GRU was pretrained on the pooled 14 rich-tier cities with per-city z-scoring (the target scarce city is never in the pretraining corpus), then fine-tuned on each of the 15 scarce-tier cities at nominal budgets of {0, 1, 10, 100}% of that city’s training window, five seeds each. Comparators on identical test folds: zero-shot Chronos-Bolt, and a LightGBM specialist refit on the same nominal budget, so the comparison is “transfer versus every strategy at the same data budget,” not “transfer versus nothing.” Two features of the smallest budgets both cut against the transfer model’s favour and are detailed in the Supplementary Information: the 0% condition uses target-city data for normalization though not for gradient updates, and the two window floors are asymmetric, so at the 1% budget the LightGBM baseline receives up to $\sim 4\times$ more data. Actual hours used are recorded per row and reported alongside nominal fractions (Supplementary Table S10). The outcome was classified against a pre-specified four-way interpretation matrix by city-level paired Wilcoxon tests, Holm-corrected across fractions, not by comparing means.

Energy and cost measurement

Primary measurement wraps each tier’s entire runner call with codecarbon⁴⁵, using a fresh tracker per call in process tracking mode at a one-second sampling interval. The metered region is the whole runner call: checkpoint load, and for trained tiers the per-fold training, as well as inference.

The two hardware components are not on equal footing. GPU energy is a hardware measurement, read from NVML counters on the study machine’s RTX 3060 Ti (the Chronos and NAS-GRU tiers). CPU energy is not: the study machine runs Windows, where the RAPL interface that would provide a hardware CPU reading is unavailable, so codecarbon derives CPU draw from a thermal-design-power model. It matches the processor (an AMD Ryzen 5 3600, 6 cores / 12 threads) to a 65 W entry in its bundled power table and scales that by the process’s measured CPU utilisation. The CPU figures are therefore modelled rather than metered.

We accordingly interpret the CPU-side numbers as comparative, directional quantities: they support relative statements between tiers measured on this one machine under an identical protocol, but do not license claims about absolute hardware energy draw or extrapolation to other processors. The GPU-side numbers, which dominate the foundation-model and NAS tiers and carry most of the headline energy contrast, are NVML-measured and stand on firmer ground. Energy is normalized to joules per 1,000 forecasts and converted to USD at 0.15 USD/kWh with PUE 1.4 (defaults varied in a sensitivity analysis, Supplementary Table S11). The supplementary TDP $\times$ latency proxy measures inference only, so it is never averaged or ratioed against a trained tier’s measured figure. Repeatability: a fixed workload was rerun five times on three cities (Beijing, Seoul, Nairobi); any city $\times$ tier cell with sd/mean $> 20\%$ is reported as a range rather than a point estimate (4 of 15 cells; Supplementary Table S8)^{35,47}.

Cost-adjusted decision rule. For each training-history regime (4–104 weeks, city-dependent) we select the tier minimizing $\text{MASE} + \lambda \cdot (\text{USD per 1,000 forecasts})$, where the cost-penalty coefficient λ is swept over {0, 500, 1500, 5000, 20000} MASE units per USD per 1,000 forecasts, producing winner maps per domain for Beijing (depth city), Seoul (rich tier), and Nairobi (scarce tier). Electricity price and PUE act on this objective as linear rescalings of λ ; the sensitivity sweep quantifies map stability.

All timestamps are handled in Coordinated Universal Time (UTC) end to end, so calendar features are computed on the UTC clock rather than each city's local time. The same convention applies to every tier and both domains, so it cannot bias any of the contrasts reported here; the full treatment is in the Supplementary Information.

Statistical framework

Hypothesis families. The confirmatory comparisons are the two Table 1 causal-covariate domain comparisons (specialist vs zero-shot foundation model on PM2.5 and on temperature) and the four E4 fine-tune budgets (transfer vs zero-shot); city-level DM significance is FDR-controlled within each domain, and E4 Wilcoxon tests are Holm-corrected across the four budgets. All other reported P values are secondary or sensitivity analyses and are read descriptively: the perfect-foresight ceilings (Table S13), the 48-hour horizon (Table S9), the panel-level sign/Wilcoxon and Friedman–Nemenyi robustness checks, the equivalence (TOST) analyses, and the data-volume correlations. We flag each as such where reported and do not apply cross-family multiplicity correction across these distinct question families.

Pairwise per-city comparisons use the Diebold–Mariano test⁵⁴ on absolute-error loss. The loss-differential series for each city is the per-step difference in absolute error, pooled over all six folds and all 24 lead times (≈ 144 values per city, one 24-step trajectory per fold origin), so it covers the same horizons that MASE averages.

We do not treat these values as independent. Errors belonging to the same forecast trajectory, to neighbouring valid times, and to different lead times are all serially correlated, so the variance of the mean loss differential is estimated with a heteroskedasticity- and autocorrelation-consistent (HAC) Newey–West estimator, using a Bartlett kernel over lags up to $h - 1 = 23$, in place of the i.i.d. variance that the original DM statistic assumes. The resulting statistic carries the Harvey–Leybourne–Newbold small-sample correction⁵⁵ and is referred to a Student $t(n - 1)$ distribution rather than a standard normal, the reference the HLN correction is derived for. Fold origins are non-overlapping, so residual cross-fold dependence reflects the series' own persistence, while within-trajectory and cross-horizon dependence is absorbed by the HAC lag window. Significant-win counts are additionally controlled with Benjamini–Hochberg FDR across the 29 cities.

Because that variance correction is nonetheless parametric, every per-city verdict was re-derived under two weaker sets of assumptions. A moving-block bootstrap resamples contiguous blocks of the loss-differential series, carrying within-block dependence through the resample rather than modelling it, with block length swept over 12, 24 and 48 steps (half a forecast trajectory, one, and two) to test rather than assume that the block spans the dependence range. A trajectory-level test then collapses each 24-step trajectory to its mean, leaving one value per non-overlapping fold origin ($n = 6$) and discarding within-trajectory and cross-lead dependence entirely. The bootstrap returns more significant cities than the HAC test in both domains, so the parametric correction is conservative rather than permissive on these data, while the deliberately underpowered trajectory-level test returns fewer, as its $n = 6$ implies (exact counts in Supplementary Table S17). Verdicts also hold lead by lead: testing each of the 24 lead times separately across cities, the sign of the mean loss differential is unchanged at 21 of 24 leads for temperature and 14 of 24 for PM2.5, and no individual lead reaches significance for temperature (3 of 24 for PM2.5), so neither domain's conclusion is an artifact of pooling across horizons. Because a non-significant test is not evidence of equivalence, the three “tie” comparisons (PM2.5 specialist vs foundation model, causal-covariate temperature specialist vs foundation model, and E4 transfer vs zero-shot) are additionally assessed with a paired two-one-sided test (TOST) against an equivalence margin of $\delta = 0.05$ MASE, at $\alpha = 0.05$.

Two conventions apply throughout: δ and α share the value 0.05 despite measuring distinct things, and TOST tests against a 90% interval rather than the conventional 95% one used for the accompanying bootstrap CI (Supplementary Information).

Because MASE is scale-free, one numerical margin does not carry one physical meaning: $\delta = 0.05$ denotes a different quantity in each domain and tier, and we state those quantities rather than leaving the margin abstract. Translating δ through each stratum's mean seasonal-naïve MAE gives, for PM2.5, $0.29 \mu\text{g m}^{-3}$ in the rich tier, $1.33 \mu\text{g m}^{-3}$ in the scarce tier and $0.83 \mu\text{g m}^{-3}$ panel-wide; and for temperature 0.18 K, 0.10 K and 0.14 K respectively. The E4 transfer comparisons are run on the 15 scarce-tier cities, so $1.33 \mu\text{g m}^{-3}$ is the figure that applies there.

Each of these is small relative to the operating thresholds of its own domain. The narrowest PM2.5 decision band at the low-concentration end of the US AQI scale spans roughly $9 \mu\text{g m}^{-3}$ and the WHO 2021 24-hour guideline value is $15 \mu\text{g m}^{-3}$, so even the scarce-tier figure is well under a sixth of the narrowest band an operator acts on. For temperature, 0.10–0.18 K is an order of magnitude below the whole-degree thresholds used in heat-health warning systems. A single δ is therefore defensible as a *strict* margin in both domains, but it is strict to differing degrees, and the scarce-tier PM2.5 case is the least strict of the three.

The verdict does depend on the margin, so we report the full sensitivity across $\delta = 0.02$, 0.05 and 0.10 together with the minimum margin at which each comparison attains equivalence (Supplementary Table S15), and state that dependence rather than presenting equivalence as margin-free. The provenance of both the equivalence analysis and the margin is documented in the Supplementary Information.

Panel-level comparisons use a binomial sign test on per-city wins and a Wilcoxon signed-rank test on paired per-city MASE;

a Friedman test with Nemenyi post-hoc across all five tiers⁵⁶ is reported as a supplementary robustness check. Uncertainty is quantified with split-conformal 95% prediction intervals^{57–59} calibrated on the first half of each series’ backtest predictions and evaluated on the second half, pooled per tier $\times$ domain (Supplementary Table S4; per-city in Supplementary Tables S6 and S7); we note that conformal exchangeability is only approximate under temporal dependence^{48,49}. The relationship between data volume and the foundation-model advantage is a Pearson correlation between usable hours and the per-city specialist-minus-FM MASE gap, with a 10,000-resample bootstrap 95% CI. Multi-seed tiers aggregate to a per-city seed-mean before any test.

Reproducibility

All canonical runs were executed under a tagged code state with per-run configuration records (`_runconfig.json`), a run registry, and SHA-256 manifests of all input data. Every quantitative claim in this manuscript is backed by a row in a results ledger mapping the claim to the artifact and exact command that produced it, and was re-verified mechanically before submission. Seeds: LightGBM 42; NAS-GRU {42–46}. The OpenAQ API key is supplied through an environment variable and never stored in code, logs, or history. Code, the analysis plan, deviations log, and fetch scripts are released with the paper.

Data availability

The datasets analysed during the current study are available in the OpenAQ repository, <https://openaq.org> (hourly PM2.5; sensor IDs for all 29 cities are listed in Supplementary Table S2), the Open-Meteo historical archive, <https://open-meteo.com> (temperature and meteorological covariates), and the UCI Machine Learning Repository, <https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data> (Beijing multi-site air-quality data). The scripts that reproduce the exact extracts used in this study, together with SHA-256 manifests of every input file, are available in the code repository referenced in the Code availability statement below.

Code availability

The full harness, analysis scripts, the analysis plan with deviations log, the per-claim results ledger, and figure/table generation code are available at <https://github.com/Muhtasim-Munif-Fahim/cost-aware-tsfm-forecasting> under the tagged release `v1.0-campaign` used for the canonical campaign.

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Author contributions

M.M.M.F.: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Visualization, Writing – Original Draft. M.R.K.: Conceptualization, Supervision, Writing – Review & Editing. Both authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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